

Dynamic Guideline: WHO-Based Child Health (IMCI) - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Scope of this WHO-based IMCI report

Integrated Management of Childhood Illness (IMCI) is WHO’s integrated child-health strategy for sick children, combining **curative and preventive** care and prioritizing the major causes of illness and death in children under 5 years: **pneumonia, diarrhoea, malaria, measles and malnutrition** [WHO IMCI activity page](#). WHO describes IMCI as focused on the **health and well-being of the child**, with the goal of reducing preventable mortality and minimizing illness [WHO IMCI team page](#).

The most current WHO IMCI-relevant clinical guideline identified in the provided evidence is:

- **Guideline Title:** *Guideline on management of pneumonia and diarrhoea in children up to 10 years of age*
- **Publication Date:** 31 December 2024
- **Target Population:** Children with pneumonia or diarrhoea up to 10 years of age
- **Scope and Objectives:** Clinical assessment, investigation, diagnosis, treatment, referral and follow-up for pneumonia and diarrhoea at community, first-level facility and hospital levels [NCBI Bookshelf introduction](#); [Guideline Central](#).

Exactly 5 WHO child-health / IMCI-related guideline records synthesized

#	WHO guideline or WHO guideline-related record	Date identified in supplied evidence	Target population	IMCI relevance	Verification notes
1	Guideline on management of pneumonia and diarrhoea in children up to 10 years of age	31 December 2024	Children up to 10 years with pneumonia or diarrhoea	Directly updates management of two leading IMCI conditions: pneumonia and diarrhoea	Uses GRADE and Evidence-to-Decision methods NCBI Bookshelf introduction ; Guideline Central

#	WHO guideline or WHO guideline-related record	Date identified in supplied evidence	Target population	IMCI relevance	Verification notes
2	WHO recommendations on child health: guidelines approved by the WHO Guidelines Review Committee	WHO reference WHO/MCA/17.08; publication date not exposed in supplied snippets	Children covered by WHO child-health recommendations	Central WHO child-health compendium relevant to IMCI	Official WHO publication; recommendation rows not extractable from supplied snippets WHO publication page
3	Recommendations for Management of Common Childhood Conditions / Pocket Book of Hospital Care for Children evidence update	GDG meeting February 2011; evidence to December 2010	Sick children requiring outpatient, first-referral or hospital care	Supports IMCI-linked management of common severe childhood conditions	Addresses pneumonia, diarrhoea, malaria, severe acute malnutrition, HIV/AIDS and life-threatening fever causes NCBI Bookshelf
4	Guideline: Assessing and Managing Children at Primary Health-Care Facilities to Prevent Overweight and Obesity in the Context of the Double Burden of Malnutrition: Updates for IMCI	2017	Infants and children presenting at primary health-care facilities	Updates IMCI nutrition assessment and counselling in the double burden of malnutrition	WHO guideline hosted on NCBI Bookshelf NCBI Bookshelf
5	WHO/UNICEF generic IMCI guidelines / IMCI chart and adaptation materials	Date not exposed in supplied snippets	Children aged 1 week up to 5 years	Core IMCI case-management algorithms for assessment, classification, treatment, counselling and follow-up	Generic materials intended for national adaptation WHO EMRO ; WHO IMCI Adaptation Guide PDF

Clarification on the requested “Meaningful youth engagement...” title

The most recent WHO publication identified with the title pattern “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**” is dated **11 December 2025**, but it concerns the **WHO Abortion Care Guideline**, not IMCI or child-health case management [WHO publications](#); [WHO event page](#). Therefore, it is **not counted among the 5 WHO IMCI-related guidelines** above. It is relevant only as a WHO guideline-development process example, particularly for stakeholder engagement.

1.2 Key Recommendations

A. Latest WHO guideline: pneumonia and diarrhoea management in children up to 10 years

The 2024 WHO guideline covers pneumonia and diarrhoea management at community, first-level facility and hospital levels. The supplied sources confirm that it uses GRADE, but they do **not** expose every recommendation’s exact WHO-assigned certainty or strength label. Therefore, the GRADE ratings below are presented as **report-assigned certainty ratings based on the evidence type available in the provided summaries**, using standard GRADE domains: risk of bias, consistency, directness, precision and publication bias.

Recommendation	GRADE Evidence Quality	Strength
Assess children with suspected pneumonia using age, respiratory signs, danger signs and, where available, pulse oximetry; when pulse oximetry is unavailable, use clinical signs of hypoxaemia such as head nodding, nasal flaring, grunting or severe tachypnoea .	Low	Strong
Treat fast-breathing pneumonia in children 2–59 months with oral amoxicillin when no severe disease or danger signs are present.	Moderate	Strong
For chest-indrawing pneumonia, give a first dose of antibiotic, commonly oral amoxicillin , and determine whether outpatient/community management or referral is appropriate based on severity, risk factors and local capacity.	Moderate	Strong
Refer or provide enhanced care for children with pneumonia and mortality risk factors; enhanced care may include hospitalization, close clinical monitoring and longer follow-up .	Low	Strong
Manage diarrhoea primarily with rehydration: oral rehydration solution for non-severe dehydration and appropriate facility-based rehydration for severe dehydration.	High	Strong
Do not use antibiotics for uncomplicated acute watery diarrhoea; reserve antibiotics for suspected dysentery or cholera according to local and WHO guidance.	Moderate	Strong

Recommendation	GRADE Evidence Quality	Strength
Provide therapeutic zinc for diarrhoea; the 2024 WHO guideline includes an updated zinc dosing recommendation, although the exact dose is not shown in the supplied snippets.	Moderate	Strong
Ensure follow-up after pneumonia or diarrhoea, especially for children with risk factors for mortality, persistent symptoms, treatment failure or poor feeding.	Low	Strong

Sources: [Guideline Central](#); [Medscape WHO guideline summary](#); [NCBI Bookshelf introduction](#); [NCBI Annex 5](#).

B. Core IMCI recommendations across the 5 WHO-related guideline records

Recommendation	GRADE Evidence Quality	Strength
Use the IMCI sequence of assess, classify, treat, counsel and follow up for sick children, rather than managing each symptom in isolation.	Low	Strong
Prioritize recognition of general danger signs and urgent referral for children with severe illness, inability to drink or breastfeed, persistent vomiting, convulsions, lethargy/unconsciousness, severe respiratory distress or severe dehydration.	Moderate	Strong
Focus first-contact care on the major IMCI conditions: pneumonia, diarrhoea, malaria, measles, malnutrition and ear infection where locally relevant.	Moderate	Strong
Adapt generic WHO/UNICEF IMCI materials to national epidemiology, drug policy, malaria transmission, HIV/TB burden, antimicrobial resistance and health-system capacity.	Low	Strong
Integrate nutrition assessment into every sick-child encounter, including wasting, growth faltering, breastfeeding/feeding problems and risk of overweight or obesity where relevant.	Moderate	Strong
Provide caregiver counselling on feeding, fluids, danger signs, medication adherence and when to return immediately.	Low	Strong
Maintain essential drugs and supplies at first-level facilities, including amoxicillin dispersible tablets, oral rehydration salts, zinc, antimalarials where indicated, vitamin A where indicated, weighing scales, respiratory timers and referral tools.	Low	Strong

Sources: [WHO IMCI activity page](#); [WHO IMCI team page](#); [WHO IMCI Adaptation Guide PDF](#); [WHO EMRO](#); [NCBI Bookshelf](#).

1.3 Guideline Development Methodology

Evidence Review Process

- The 2024 WHO pneumonia and diarrhoea guideline used the **GRADE** approach to assess certainty of evidence and support recommendation development [NCBI Bookshelf introduction](#).
- GRADE factors included:
 - certainty of evidence;
 - benefits and harms;
 - values and preferences;
 - resource use;
 - feasibility and implementation considerations [NCBI Bookshelf introduction](#).
- The NCBI annex confirms GRADE tables for specific pneumonia and diarrhoea PICO questions, including:
 - oral amoxicillin versus no antibiotic for fast-breathing pneumonia;
 - outpatient oral amoxicillin versus inpatient injectable antibiotics for chest-indrawing pneumonia;
 - antibiotics versus no antibiotics for acute watery or persistent diarrhoea;
 - oral zinc versus no zinc, including dose, duration and formulation [NCBI Annex 5](#).

Consensus Method

- The 2024 guideline used an **Evidence-to-Decision framework** during the guideline meeting [NCBI Bookshelf introduction](#).
- Earlier WHO IMCI and common-childhood-condition recommendations used guideline development groups and evidence review processes; for example, the WHO GDG meeting for common childhood conditions was held in **February 2011**, using evidence available as of **December 2010** [NCBI Bookshelf](#).

Conflict of Interest Management

- The supplied summaries confirm WHO guideline methods and GRADE use for the 2024 pneumonia and diarrhoea guideline, but do **not** expose detailed conflict-of-interest declarations.
- As a WHO guideline, it would be expected to follow WHO guideline-development procedures, but itemized conflict-of-interest management cannot be independently extracted from the supplied snippets.

GRADE interpretation used in this report

GRADE category	Interpretation for this report
High	Further research is unlikely to change confidence in the estimate or clinical direction.
Moderate	Further research may change confidence or refine magnitude, but the recommendation is clinically stable.
Low	Further research is likely to affect confidence; recommendation relies on indirect, implementation or older evidence, or strong clinical rationale.
Very Low	Evidence is very uncertain, indirect, sparse or primarily observational/programmatic.

GRADE category	Interpretation for this report
Strong recommendation	Most children should receive the recommended action; benefits clearly outweigh harms or the action is essential for safety.
Conditional recommendation	Action should be individualized by context, feasibility, patient risk, caregiver preference or resource availability.

II. Evidence Summary After WHO Latest Guideline Release (WHO recommendations on child health: guidelines approved by the WHO Guidelines Review Committee) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

The evidence window for “new evidence” was defined as:

- **Start date:** 01 January 2025, immediately after the 31 December 2024 WHO pneumonia and diarrhoea guideline publication.
- **Evidence cutoff date:** 29 April 2026.
- **Clinical focus:** IMCI, IMNCI, pneumonia, diarrhoea, malaria, measles, malnutrition, digital IMCI/eIMCI, integrated community case management, iCCM and under-five mortality.

Representative searches included:

- 2025 2026 IMCI pneumonia diarrhoea eIMCI iCCM under-five mortality systematic review trial implementation
- IMCI systematic review meta-analysis ICT 2024
- WHO 2024 pneumonia diarrhoea children up to 10 GRADE recommendations
- WHO IMCI child health guideline GRC pneumonia diarrhoea malaria malnutrition measles newborn

Databases and Sources Searched

The supplied research summaries included WHO and WHO-linked sources, NCBI Bookshelf, Guideline Central, Medscape, Cochrane, DARE/NCBI, ScienceDirect, Clinical Microbiology and Infection, Save the Children implementation material, and related academic pages.

Key sources included:

- WHO IMCI pages [WHO IMCI team page](#)
- WHO child-health compendium [WHO publication page](#)
- 2024 WHO pneumonia/diarrhoea guideline summaries [Guideline Central](#); [NCBI Bookshelf introduction](#)
- Cochrane IMCI review [Cochrane](#)
- Digital/ICT IMCI review [PMC](#)
- Community-based pneumonia/diarrhoea review [NCBI Bookshelf/DARE](#)

Inclusion Criteria

Included evidence was required to be:

- 1. relevant to IMCI, child pneumonia, diarrhoea, malnutrition, malaria, measles, iCCM, digital IMCI or under-five mortality;
- 2. published or updated after 31 December 2024, where possible;
- 3. from WHO, authoritative institutions, systematic reviews, randomized trials, implementation evaluations or guideline registries;
- 4. clinically relevant to diagnosis, treatment or management.

Exclusion Criteria

Excluded or downgraded evidence included:

- non-clinical process documents unless relevant to guideline development;
- sources without extractable findings;
- non-WHO educational materials when official WHO recommendation data were unavailable;
- country-specific implementation documents when they could not be generalized globally;
- preprints or abstracts without sufficient clinical data, unless clearly labelled as low-certainty supporting context.

Overall finding for post-guideline evidence

Within the supplied evidence, **no clearly eligible post-31 December 2024 clinical trial, systematic review or implementation study with extractable effect estimates was identified** that would overturn the 2024 WHO pneumonia/diarrhoea guidance or core IMCI recommendations. The most relevant new or recent evidence remains:

- 1. the 2024 WHO pneumonia/diarrhoea guideline itself;
- 2. the digital/ICT IMCI systematic review searching literature through February 2024;
- 3. older Cochrane and DARE evidence supporting IMCI/iCCM implementation;
- 4. programmatic digital IMNCI implementation evidence from Kenya.

2.2 Key New Studies

A. Post-2024 evidence table

Study / Source	Design	Key Findings	GRADE Quality
Post-31 December 2024 IMCI clinical effectiveness studies identified in supplied evidence	None with extractable outcomes	No eligible post-2024 RCT, systematic review or implementation evaluation with extractable effect estimates was found in the supplied summaries.	Very Low

Study / Source	Design	Key Findings	GRADE Quality
WHO “Meaningful youth engagement...” case study, 11 December 2025	Guideline-development case study, not clinical effectiveness evidence	Demonstrates youth consultations and youth-led convening in the 2022 WHO Abortion Care Guideline process, but it is not an IMCI guideline and does not change child-health management recommendations.	Very Low for IMCI applicability
2025 Scientific Reports IMCI-related article signal	Observational/contextual literature signal	Characterizes WHO/UNICEF IMCI as a strategy for identifying and treating illnesses in children under 5 in LMICs; no extractable IMCI treatment effect provided in the supplied summary.	Very Low

Sources: [WHO publications](#); [WHO event page](#); [Scientific Reports](#).

B. Recent but mostly pre-2025 evidence relevant to implementation

Although not published after the WHO 2024 guideline, the following evidence remains relevant for implementation and contextual interpretation.

Study / Source	Design	Key Findings	GRADE Quality
ICT to enhance IMCI/iCCM implementation	Systematic review and meta-analysis; searched MEDLINE, EMBASE, Cochrane Library and grey literature from January 2010 to February 2024	Focused on digital/ICT tools for IMCI and iCCM implementation; supplied snippets do not provide pooled effect estimates.	Low
Cochrane review of IMCI for children under 5	Systematic review of trials; search to 2015	IMCI may reduce child mortality. Bangladesh trial: RR 0.87, 95% CI 0.68–1.10. Pooled IMCI/IMNCI mortality: RR 0.85, 95% CI 0.78–0.93.	Low
IMNCI India evidence in Cochrane review	Cluster-randomized evidence summarized in Cochrane	Infant mortality lower with IMNCI: HR 0.85, 95% CI 0.77–0.94; neonatal mortality estimate compatible with little or no effect: HR 0.91, 95% CI 0.80–1.03.	Low

Study / Source	Design	Key Findings	GRADE Quality
IMCI nutrition outcomes in Cochrane evidence	Systematic review evidence	Stunting: RR 0.94, 95% CI 0.84–1.06; wasting: RR 1.04, 95% CI 0.87–1.25.	Low for stunting; Moderate for wasting
Community-based pneumonia/diarrhoea interventions	Systematic review in DARE; 24 studies	Increased care-seeking for pneumonia by 13%, diarrhoea by 9%; ORS use increased 160%; zinc use increased 80%; unnecessary antibiotics for diarrhoea declined 75%; pneumonia treatment failure decreased 40%; pneumonia-specific mortality reduced 32%; diarrhoea mortality evidence weak.	Low
Kenya digital IMNCI/THINKMD pilot	Programmatic implementation evaluation	Provider preference for amoxicillin dispersible tablets increased from 3% to 38%; knowledge that antibiotics should not be used for simple diarrhoea increased from 14% to 50%; >90% provider compliance with IMNCI reported at endline.	Very Low

Sources: [ICT systematic review and meta-analysis, PMC](#); [Cochrane](#); [ResearchGate record of Cochrane review](#); [NCBI Bookshelf/DARE](#); [Save the Children fact sheet PDF](#).

2.3 Evidence Gaps and Future Research Needs

Major evidence gaps

1. Post-2024 clinical effectiveness evidence

- No post-31 December 2024 RCT or systematic review with extractable IMCI outcome data was identified in the supplied evidence.
- Priority outcomes: all-cause under-five mortality, pneumonia mortality, diarrhoea mortality, hospitalization, referral completion, treatment failure and adverse events.

2. Digital IMCI / eIMCI effectiveness

- Digital IMCI tools are increasingly used, and a systematic review exists, but supplied snippets do not provide pooled effect estimates.
- Future studies should measure:
 - diagnostic accuracy;
 - antibiotic appropriateness;
 - ORS/zinc use;

- referral completion;
- mortality or severe morbidity;
- cost-effectiveness.

3. **Pneumonia diagnosis without pulse oximetry**

- WHO identifies clinical signs for hypoxaemia when pulse oximetry is unavailable.
- More evidence is needed on performance of these signs across settings, altitude, age groups, malnutrition and HIV exposure.

4. **Management of high-risk children**

- Children with malnutrition, HIV exposure/infection, prematurity, incomplete immunization, recurrent illness or social vulnerability require clearer risk-stratified algorithms.

5. **Antimicrobial stewardship**

- IMCI must preserve access to lifesaving antibiotics for pneumonia and dysentery while reducing inappropriate antibiotic use for viral respiratory illness and simple watery diarrhoea.

6. **Nutrition and double burden of malnutrition**

- IMCI historically emphasized undernutrition, but the 2017 WHO update adds overweight/obesity prevention in primary care.
- More evidence is needed on integrating wasting, stunting, micronutrient deficiency and overweight risk in one practical algorithm.

7. **Equity, caregiver engagement and adolescent/youth participation**

- The WHO youth-engagement case study is not IMCI-specific, but it highlights a broader methodological gap: child-health guideline development should more systematically include caregivers, adolescents, youth advocates and community health workers where appropriate [WHO publications](#).

2.4 Updated Recommendations Based on New Evidence

Because no post-2024 clinical evidence with extractable effect estimates was identified that changes WHO IMCI-related recommendations, the current report supports **implementation reinforcement rather than substantive recommendation change**.

Original Recommendation	New Evidence Impact	Updated Recommendation
Use IMCI to assess, classify, treat, counsel and follow up sick children.	No post-2024 evidence identified that changes this approach. WHO continues to frame IMCI as a broad child-survival strategy.	Continue IMCI as the standard integrated first-contact child-health approach, adapted to national epidemiology and resources.

Original Recommendation	New Evidence Impact	Updated Recommendation
Treat childhood pneumonia according to severity, using oral amoxicillin for eligible non-severe cases and referral/enhanced care for severe or high-risk disease.	2024 WHO guideline reinforces pneumonia management; no newer contradictory evidence found.	Maintain WHO 2024 pneumonia recommendations; strengthen oxygen assessment, pulse oximetry access and referral systems.
Use clinical signs of hypoxaemia when pulse oximetry is unavailable.	WHO 2024 guidance identifies signs including head nodding, nasal flaring, grunting and severe tachypnoea. No newer evidence changes this.	Use pulse oximetry whenever available; if unavailable, treat hypoxaemia signs as high-risk and refer or manage urgently.
Manage diarrhoea with rehydration, zinc and selective antibiotics only when indicated.	2024 WHO guideline includes updated zinc dosing; no newer evidence changes the core approach.	Continue ORS/rehydration and zinc; avoid antibiotics for uncomplicated watery diarrhoea; use antibiotics for suspected dysentery or cholera per guidance.
Adapt IMCI to local policy, medicines and epidemiology.	Digital and implementation evidence supports the importance of adherence tools, but certainty remains low to very low.	Countries may use digital IMCI tools if validated, monitored and aligned with WHO/national algorithms; implementation should include audit and safety monitoring.
Include nutrition assessment in sick-child care.	2017 WHO IMCI update remains relevant; no newer evidence overturns it.	Assess all children for wasting, feeding problems and growth concerns; include overweight/obesity prevention counselling where contextually relevant.
Use caregiver counselling and follow-up.	Evidence remains indirect but clinically important.	Provide clear return instructions for danger signs, feeding/fluid advice, medication adherence and scheduled follow-up.

Practical IMCI Algorithm for Current Use

1. Initial assessment: all sick children

Assess immediately for:

- inability to drink or breastfeed;
- vomiting everything;
- convulsions;

- lethargy or unconsciousness;
- severe respiratory distress;
- cyanosis or suspected hypoxaemia;
- severe dehydration;
- severe acute malnutrition;
- very young age or high-risk comorbidity.

Action: Any general danger sign should trigger urgent treatment, pre-referral therapy where appropriate and referral or hospital-level care.

2. Pneumonia

- Count respiratory rate by age.
- Look for chest indrawing, grunting, nasal flaring, head nodding, cyanosis and inability to feed.
- Use pulse oximetry where available.
- If pulse oximetry is unavailable, treat **head nodding, nasal flaring, grunting or severe tachypnoea** as possible hypoxaemia signs [Medscape WHO guideline summary](#).

Management:

- Non-severe fast-breathing pneumonia: oral amoxicillin when eligible.
- Chest-indrawing pneumonia: oral amoxicillin may be used in selected settings, but assess need for referral.
- Severe pneumonia or high-risk child: oxygen if needed, antibiotics, urgent referral/hospitalization and close monitoring.

3. Diarrhoea

Assess:

- duration;
- blood in stool;
- cholera risk;
- dehydration severity;
- ability to drink;
- persistent vomiting;
- malnutrition.

Management:

- No/some dehydration: ORS and continued feeding.
- Severe dehydration: urgent facility-based rehydration.
- Zinc: provide therapeutic zinc according to current WHO/national dose.
- Antibiotics: avoid in uncomplicated watery diarrhoea; use for suspected dysentery or cholera according to guidance.

4. Malnutrition and feeding

- Weigh the child and assess growth/nutritional status.
- Screen for severe acute malnutrition.

- Assess breastfeeding, complementary feeding, appetite and illness-related feeding problems.
- Counsel caregivers on continued feeding and extra fluids during illness.
- Apply the 2017 WHO IMCI update for prevention of overweight and obesity where relevant [NCBI Bookshelf](#).

5. Follow-up and caregiver counselling

Give caregivers clear instructions on:

- danger signs requiring immediate return;
- medication dose and duration;
- ORS preparation and administration;
- continued feeding and breastfeeding;
- follow-up timing;
- referral destination if needed.

Overall Conclusion

As of **29 April 2026**, the latest WHO IMCI-relevant clinical guideline identified is the **31 December 2024 WHO guideline on management of pneumonia and diarrhoea in children up to 10 years of age**. It reinforces core IMCI principles: structured assessment, severity classification, appropriate antibiotics for pneumonia, rehydration and zinc for diarrhoea, selective antibiotic use, referral for severe disease and follow-up.

No post-2024 clinical evidence with extractable effect estimates was identified in the supplied summaries that would require changing WHO-based IMCI recommendations. The most important current action is therefore **high-quality implementation**: ensure trained health workers, reliable essential medicines, pulse oximetry and oxygen where possible, safe referral pathways, caregiver counselling, and locally adapted IMCI algorithms aligned with the 2024 WHO pneumonia/diarrhoea guideline and other WHO child-health recommendations.